

TrES-3b

R-0358

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230615 · created 2026-07-18T12:06:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460110.7717 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.012
Transit depth (Rp/Rs)^2	2.93 +/- 0.42 [%]
Orbital Inclination (inc)	82.69 +/- 0.38
Airmass coefficient 1 (a1)	1.0246 +/- 0.0089
Airmass coefficient 2 (a2)	-0.021 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#3 - [99, 304]
Optimal Method	PSF photometry
Transit Duration (day)	0.0627 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.171 ± 0.012 vs 0.16309 (deviation 0.57σ)
Duration fit vs published	0.0627 d vs 0.0567 d (deviation 1.7σ)
Mid-time O–C	-1.4 min
Depth SNR	12.4
Residual scatter	0.86 %

Light curve

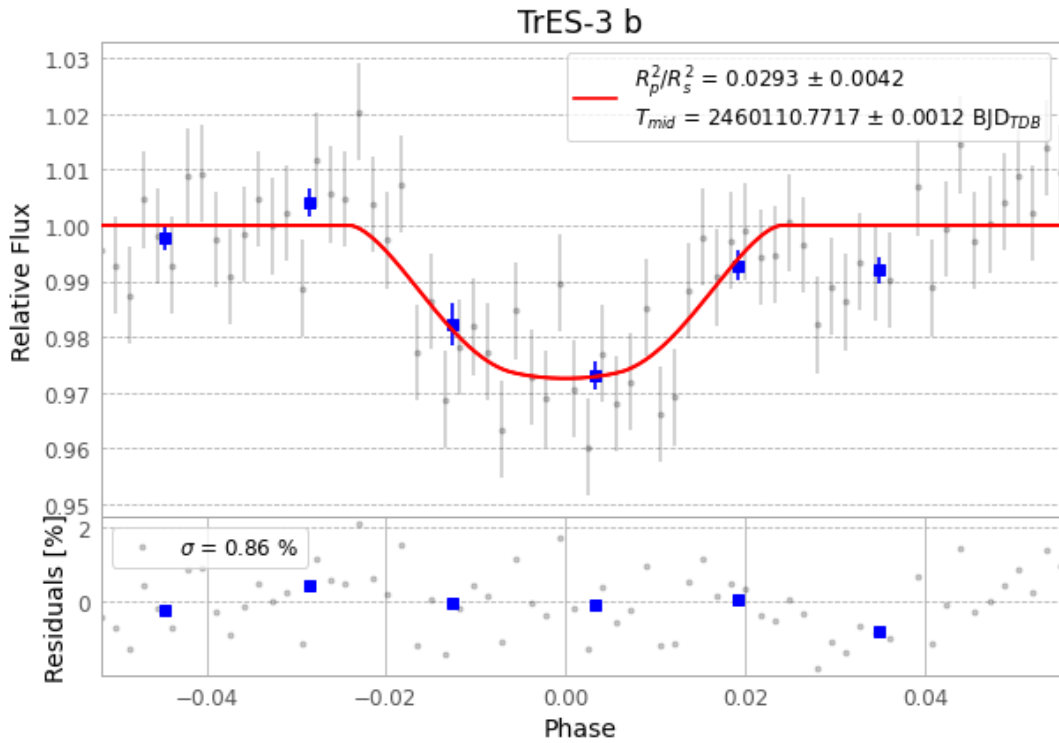


Figure 1 — FinalLightCurve_TrES-3 b_2023-06-14.png (SHA-256 79104ee6f1ea7aef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52eb0d51bb283dd3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

68 FITS frames (45 MB), TRES-3230615044815.FITS → TRES-3230615080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230615.log (8d9476e05db3...), FinalLightCurve_TrES-3 b_2023-06-14.png (79104ee6f1ea...), FinalLightCurve_TrES-3 b_2023-06-14.csv (ebf2902fae0a...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 12:06Z.